CHAPTER 2

Introduction to Linear Programming

Linear programming (LP) applies to optimization models in which the objective and constraint functions are strictly linear. The technique is used in a wide range of applications, including agriculture, industry, transportation, economics, health systems, behavioral and social sciences, and the military. It also boasts efficient computational algorithms for problems with thousands of constraints and variables. Indeed, because of its tremendous computational efficiency, LP forms the backbone of the solution algorithms for other OR models, including integer, stochastic, and nonlinear programming.

This chapter starts with the case of a two-variable model and presents its graphical solution. The graphical solution provides insight into the development of the general algebraic *simplex method* (see Chapter 3). It also gives concrete ideas for the development and interpretation of sensitivity analysis in LP. The chapter closes with the formulation and solution interpretation of a number of realistic applications.

2.1 TWO-VARIABLE LP MODEL

This section deals with the graphical solution of a two-variable LP. Though two-variable problems hardly exist in practice, the treatment provides concrete ideas for the development of the general solution algorithm presented in Chapter 3.

Example 2.1-1 (The Reddy Mikks Company)

Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		*Maximum daily availability (tons)*
	Exterior paint	*Interior paint*	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton ($1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that of exterior paint by more than 1 ton. Also, the maximum daily demand of interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

The LP model, as in any OR model, has three basic components.

1. Decision **variables** that we seek to determine
2. **Objective** (goal) that we aim to optimize
3. **Constraints** that we need to satisfy

The proper definition of the decision variables is an essential first step in the development of the model. Once done, the task of constructing the objective function and the constraints is more straightforward.

For the Reddy Mikks problem, we need to determine the amounts to be produced of exterior and interior paints. Thus the variables of the model are defined as

x_1 = Tons produced daily of exterior paint

x_2 = Tons produced daily of interior paint

To construct the objective function, the company wants to increase its profit as much as possible. Letting z represent the total daily profit (in thousands of dollars), the objective of the company is expressed as

$$\text{Maximize } z = 5x_1 + 4x_2$$

Next, we construct the constraints that restrict raw materials usage and demand. The raw materials restrictions are expressed verbally as

$$\begin{pmatrix}\text{Usage of a raw material}\\ \text{by both paints}\end{pmatrix} \le \begin{pmatrix}\text{Maximum raw material}\\ \text{availability}\end{pmatrix}$$

From the data of the problem, we have

$$\text{Usage of raw material } M1 \text{ per day} = 6x_1 + 4x_2 \text{ tons}$$

$$\text{Usage of raw material } M2 \text{ per day} = 1x_1 + 2x_2 \text{ tons}$$

Because the daily availabilities of raw materials $M1$ and $M2$ are limited to 24 and 6 tons, respectively, the associated restrictions are given as

$$6x_1 + 4x_2 \leq 24 \quad \text{(Raw material } M1\text{)}$$

$$x_1 + 2x_2 \leq 6 \quad \text{(Raw material } M2\text{)}$$

The first demand restriction says that the difference between the daily production of interior and exterior paints, $x_2 - x_1$, does not exceed 1 ton, which translates to $x_2 - x_1 \leq 1$. The second demand restriction stipulates that the maximum daily demand of interior paint is limited to 2 tons, which translates to $x_2 \leq 2$.

An implicit (or "understood-to-be") restriction is that variables x_1 and x_2 cannot assume negative values. The **nonnegativity restrictions**, $x_1 \geq 0, x_2 \geq 0$, account for this requirement.

The complete Reddy Mikks model is

$$\text{Maximize } z = 5x_1 + 4x_2$$

subject to

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

Any values of x_1 and x_2 that satisfy all the constraints of the model constitute a **feasible solution**. For example, the solution, $x_1 = 3$ tons per day and $x_2 = 1$ ton per day, is feasible because it does not violate any of the constraints, including the nonnegativity restrictions. To verify this result, substitute $(x_1 = 3, x_2 = 1)$ in the left-hand side of each constraint. For example, in the first constraint, $6x_1 + 4x_2 = 6 \times 3 + 4 \times 1 = 22$, which is less than the right-hand side of the constraint (= 24). The value of the objective function associated with the solution $(x_1 = 3, x_2 = 1)$ is $z = 5 \times 3 + 4 \times 1 = 19$ (thousand dollars).

From the standpoint of the entire model, we are interested in finding the **optimum feasible solution** that yields the maximum total profit while satisfying all the constraints. Enumeration of the feasible solutions is not acceptable because the model has an infinite number of feasible solutions. Instead, we need a systematic procedure that will locate the optimum solution efficiently. The graphical method in Section 2.3 and its algebraic generalization in Chapter 3 address this point.

In the preceding example, the objective and constraints functions are all linear. **Linearity** implies that the LP must satisfy two properties: **proportionality** and **additivity**.

1. *Proportionality* requires the contribution of each decision variable in the objective function and its requirements in the constraints to be *directly proportional* to the value of the variable. For example, in the Reddy Mikks model, the quantities $5x_1$ and $4x_2$ give the profits for producing x_1 and x_2 tons of exterior and interior paint, respectively, with the unit profits per ton, 5 and 4, providing the constants of proportionality. If, on the other hand, Reddy Mikks grants some sort of quantity discounts when sales exceed certain amounts, then the profit will no longer be proportional to the production amounts, x_1 and x_2.

2. *Additivity* stipulates that the total contribution of all the variables in the objective function and their requirements in the constraints are the direct sum of the individual contribution or requirements of each variable. In the Reddy Mikks model, the total profit equals the sum of the two individual profit components. If, however, the two products *compete* for the same market share in a manner that an increase in sales of one adversely affects the other, then the additivity property is not satisfied.

PROBLEM SET 2.1A

1. For the Reddy Mikks model, construct each of the following constraints and express them with a constant right-hand side:
   (a) The daily demand for interior paint exceeds that of exterior paint by *at least* 1 ton.
   (b) The daily usage of raw material *M2* is *at most* 6 tons and *at least* 3 tons.
   (c) The demand for interior paint cannot be less than the demand for exterior paint.
   (d) The minimum quantity that should be produced of both the interior and the exterior paint is 3 tons.
   (e) The proportion of interior paint to the total production of both interior and exterior paints must not exceed .5.
2. Determine the best *feasible* solution among the following (feasible and infeasible) solutions of the Reddy Mikks model:
   (a) $x_1 = 1, x_2 = 4$
   (b) $x_1 = 2, x_2 = 2$
   (c) $x_1 = 3, x_2 = 1.5$
   (d) $x_1 = 2, x_2 = 1$
   (e) $x_1 = 2, x_2 = -1$
3. For the feasible solution $x_1 = 2, x_2 = 2$ of the Reddy Mikks model, determine
   (a) The unused amount of raw material *M1*.
   (b) The unused amount of raw material *M2*.
4. Suppose that Reddy Mikks sells its exterior paint to a wholesaler at quantity discount. The profit per ton is \$5000 if the contractor buys no more than 2 tons daily and \$4500 otherwise. Can this situation be modeled as an LP model?

2.2 GRAPHICAL LP SOLUTION

The graphical procedure includes two steps:

1. Determination of the solution space that defines *all* feasible solutions of the model.
2. Determination of the optimum solution from among all the feasible points in the solution space.

The procedure uses two examples to show how the maximization and minimization objective functions are handled.

2.2.1 Solution of a Maximization Model

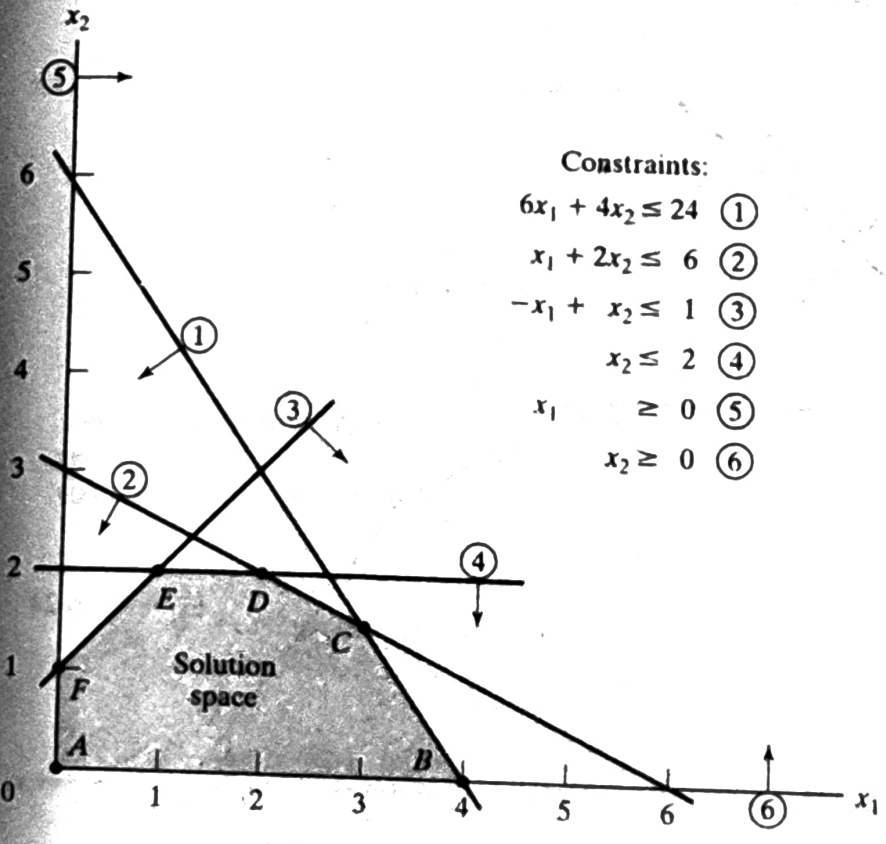
Example 2.2-1

This example solves the Reddy Mikks model of Section 2.1.

Step 1. *Determination of the Feasible Solution Space:*
First, we account for the nonnegativity constraints $x_1 \geq 0$ and $x_2 \geq 0$. In Figure 2.1, the horizontal axis x_1 and the vertical axis x_2 represent the exterior- and interior-paint variables, respectively. It then follows that the nonnegativity constraints will restrict the solution space area to the first quadrant that lies above the x_1-axis and to the right of the x_2-axis.

To account for the remaining four constraints, first replace each inequality with an equations and then graph the resulting straight line by locating two distinct points on it. For example, after replacing $6x_1 + 4x_2 \leq 24$ with the straight line $6x_1 + 4x_2 = 24$, two distinct points can be determined first by setting $x_1 = 0$ to obtain $x_2 = \frac{24}{4} = 6$ and then setting $x_2 = 0$ to obtain $x_1 = \frac{24}{6} = 4$. Thus, the line passes through the two points $(0, 6)$ and $(4, 0)$, as shown by line (1) in Figure 2.1.

Next, consider the effect of the inequality. All the inequality does is divide the (x_1, x_2)-plane into two half-spaces, one on each side of the plotted line. Only one of these two halves satisfies the inequality. To determine the correct side, choose any *reference point* in the first quadrant. If it satisfies the inequality, then the side in which it lies is the feasible half-space. Else, the other side is. It is convenient computationally to select $(0, 0)$ as the reference



Constraints:

$6x_1 + 4x_2 \leq 24$ (1)
$x_1 + 2x_2 \leq 6$ (2)
$-x_1 + x_2 \leq 1$ (3)
$x_2 \leq 2$ (4)
$x_1 \geq 0$ (5)
$x_2 \geq 0$ (6)

FIGURE 2.1
Feasible space of the Reddy Mikks model

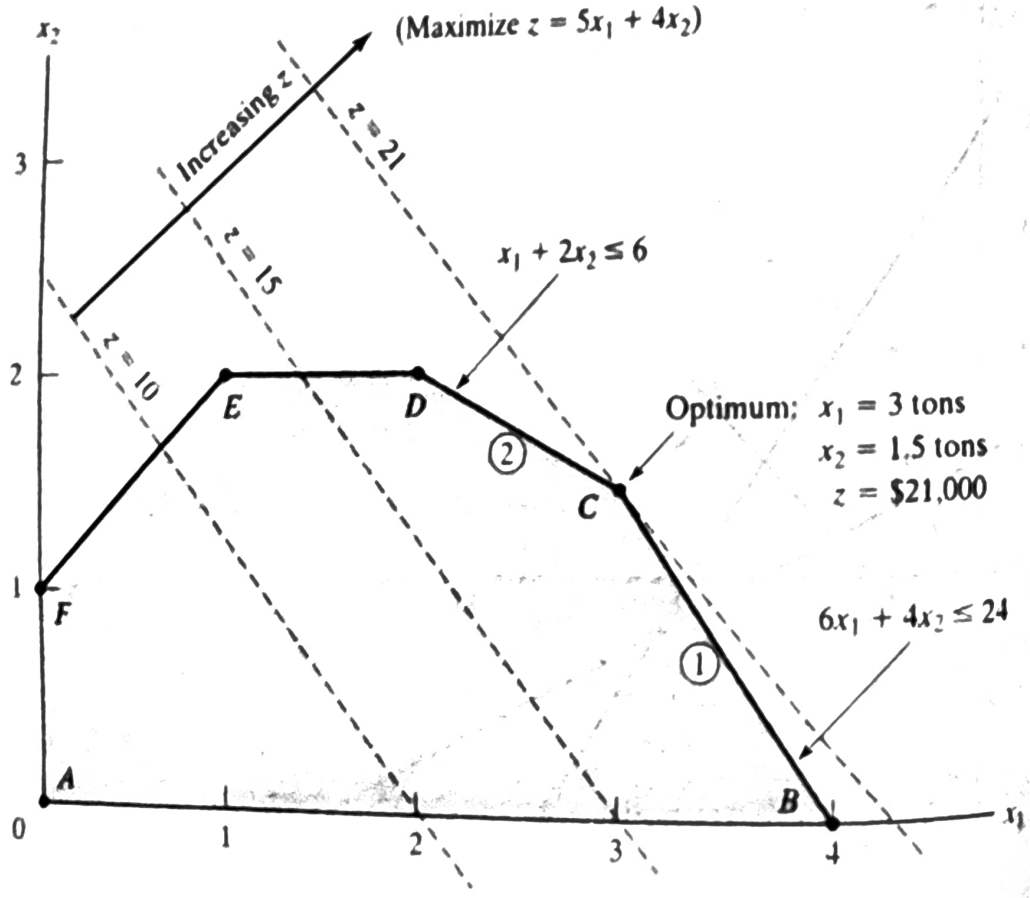
point, unless the line happens to pass through the origin, in which case another point must be selected.

The use of the reference point (0, 0) is illustrated with the constraint $6x_1 + 4x_2 \leq 24$. Because $6 \times 0 + 4 \times 0 = 0$ is less than 24, the half-space representing the inequality includes the origin (as shown by the arrow in Figure 2.1). To demonstrate the use of other reference points, consider (6, 0). Here, $6 \times 6 + 4 \times 0 = 36$, which is larger than the right-hand side of the first constraint, meaning that the side in which (6, 0) lies is not feasible for the inequality. This result is consistent with the one obtained using (0, 0) as the reference point.

Application of the reference point procedure to all the constraints of the model produces the feasible space shown in Figure 2.1.

Step 2. *Determination of the Optimum Solution:*
The feasible space in Figure 2.1 is delineated by the line segments joining the corner points *A*, *B*, *C*, *D*, *E*, and *F*. Any point within or on the boundary of the space *ABCDEF* is feasible, in the sense that it satisfies all the constraints. Because the feasible space *ABCDEF* consists of an *infinite* number of points, the need for a systematic procedure to identify the optimum solution is obvious.

The determination of the optimum solution requires identifying the direction in which the profit function $z = 5x_1 + 4x_2$ increases (recall that we are *maximizing* z). We can do so by assigning *arbitrary* increasing values to z. For example, using $z = 10$ and $z = 15$ would be equivalent to graphing the two lines $5x_1 + 4x_2 = 10$ and $5x_1 + 4x_2 = 15$. Thus, the direction of increase in z is as shown Figure 2.2. The optimum solution occurs at *C*, which is the point in the solution space beyond which any further increase in z will put us outside the boundaries of *ABCDEF*.



IGURE 2.2
ptimum solution of the
eddy Mikks model

The values of x_1 and x_2 associated with the optimum point C are determined by solving the equations associated with lines (1) and (2)—that is,

$$6x_1 + 4x_2 = 24$$

$$x_1 + 2x_2 = 6$$

The solution is $x_1 = 3$ and $x_2 = 1.5$ with $z = 5 \times 3 + 4 \times 1.5 = 21$. This calls for a daily product mix of 3 tons of exterior paint and 1.5 tons of interior paint. The associated daily profit is \$21,000.

It is not accidental that the optimum solution occurs at a **corner point** of the solution space where two lines intersect. Indeed, if we change the slope of the profit function z (by changing its coefficients), we will discover that the optimum solution always occurs at one of these *corner* points. This observation is key to the development of the general *simplex algorithm* presented in Chapter 3.

PROBLEM SET 2.2A

1. Determine the feasible space for each of the following independent constraints, given that $x_1, x_2 \geq 0$.
   (a) $-3x_1 + x_2 \leq 6$
   (b) $x_1 - 2x_2 \geq 5$
   (c) $2x_1 - 3x_2 \leq 12$
   (d) $x_1 - x_2 \leq 0$
   (e) $-x_1 + x_2 \geq 0$
2. Identify the direction of increase in z in each of the following cases:
   (a) Maximize $z = x_1 - x_2$
   (b) Maximize $z = -5x_1 - 6x_2$
   (c) Maximize $z = -x_1 + 2x_2$
   (d) Maximize $z = -3x_1 + x_2$
3. Determine the solution space and the optimum solution of the Reddy Mikks model for each of the following independent changes:
   (a) The maximum daily demand for exterior paint is at most 2.5 tons.
   (b) The daily demand for interior paint is at least 2 tons.
   (c) The daily demand for interior paint is exactly 1 ton higher than that for exterior paint.
   (d) The daily availability of raw material $M1$ is at least 24 tons.
   (e) The daily availability of raw material $M1$ is at least 24 tons, and the daily demand for interior paint exceeds that of exterior paint by at least 1 ton.
4. For the (original) Reddy Mikks model, identify the *corner* point(s) that define the optimum solution for each of the following objective functions:
   (a) $z = 3x_1 + x_2$
   (b) $z = x_1 + 3x_2$
   (c) $z = 6x_1 + 4x_2$

   How does the solution in (c) differ from those in (a) and (b)?

5. Jack is an aspiring freshman at Ulern University. He realizes that "all work and no play make Jack a dull boy." As a result, Jack wants to apportion his available time of about 10 hours a day between work and play. He estimates that play is twice as much fun as work. He also wants to study at least as much as he plays. However, Jack realizes that if he is going to get all his homework assignments done, he cannot play more than 4 hours a day. How should Jack allocate his time to maximize his pleasure from both work and play?

2.2.2 Solution of a Minimization Model

Example 2.2-2 (Diet Problem)

Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions:

	lb per lb of feedstuff		
Feedstuff	*Protein*	*Fiber*	*Cost ($/lb)*
Corn	.09	.02	.30
Soybean meal	.60	.06	.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber. Ozark Farms wishes to determine the daily minimum-cost feed mix.

Because the feed mix consists of corn and soybean meal, the decision variables of the model are defined as

x_1 = lb of corn in the daily mix

x_2 = lb of soybean meal in the daily mix

The objective function seeks to minimize the total daily cost (in dollars) of the feed mix and is thus expressed as

$$\text{minimize } z = .3x_1 + .9x_2$$

The constraints of the model reflect the daily amount needed and the dietary requirements. Because Ozark Farms needs at least 800 lb of feed a day, the associated constraint can be expressed as

$$x_1 + x_2 \geq 800$$

As for the protein dietary requirement constraint, the amount of protein included in x_1 lb of corn and x_2 lb of soybean meal is $(.09x_1 + .6x_2)$ lb. This quantity should equal at least 30% of the total feed mix $(x_1 + x_2)$ lb; that is

$$.09x_1 + .6x_2 \geq .3(x_1 + x_2)$$

In a similar manner, the fiber constraint is constructed as

$$.02x_1 + .06x_2 \leq .05(x_1 + x_2)$$

The constraints are simplified by grouping all the terms in x_1 and x_2 and moving them to the left-hand side of each inequality, leaving only a constant on the right-hand side. The complete model thus becomes

$$\text{minimize } z = .3x_1 + .9x_2$$

subject to

$$x_1 + x_2 \geq 800$$
$$.21x_1 - .30x_2 \leq 0$$
$$.03x_1 - .01x_2 \geq 0$$
$$x_1, x_2 \geq 0$$

Figure 2.3 provides the graphical solution of the model. Unlike those of the Reddy Mikks model (Example 2.2-1), the second and third constraints pass through the origin. To plot the associated straight lines, we need one additional point, which can be obtained by assigning one of the variables a value and then solving for the other variable. For example, in the second constraint, $x_1 = 200$ will yield $.21 \times 200 - .3x_2 = 0$, or $x_2 = 140$. This means that the straight line $.21x_1 - .3x_2 = 0$ passes through (0, 0) and (200, 140). Note also that (0, 0) cannot be used as a reference point in constraints 2 and 3 because both lines pass through the origin. Instead, any other point [e.g., (100, 0) or (0, 100)] can be used for that purpose.

Because the present model seeks the minimization of the objective function, we need to reduce the value of z as much as possible in the direction shown in Figure 2.3. The optimum solution is the intersection of the two lines $x_1 + x_2 = 800$ and $.21x_1 - .3x_2 = 0$, which yields $x_1 = 470.6$ lb and $x_2 = 329.4$ lb. The associated minimum cost of the feed mix is $z = .3 \times 470.6 + .9 \times 329.4 = \$437.64$ per day.

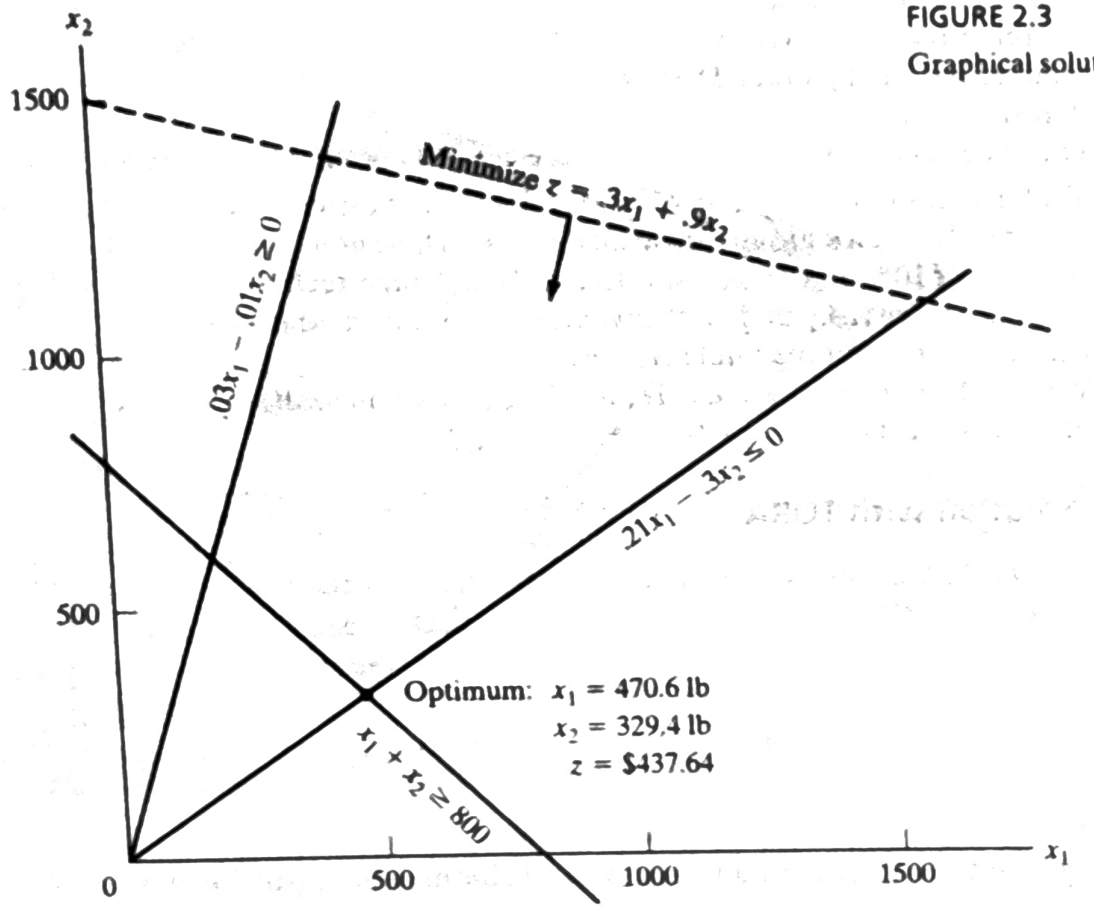


FIGURE 2.3
Graphical solution of the diet model

PROBLEM SET 2.2B

1. Identify the direction of decrease in z in each of the following cases:
   (a) Minimize $z = 4x_1 - 2x_2$
   (b) Minimize $z = -3x_1 + x_2$
   (c) Minimize $z = -x_1 - 2x_2$
2. For the diet model, suppose that the daily availability of corn is limited to 450 lb. Identify the new solution space, and determine the new optimum solution.
3. For the diet model, what type of optimum solution would the model yield if the feed mix should not exceed 800 lb a day? Does the solution make sense?
4. John must work at least 20 hours a week to supplement his income while attending school. He has the opportunity to work in two retail stores: in store 1, John can work between 5 and 12 hours a week, and in store 2 he is allowed to work between 6 and 10 hours. Both stores pay the same hourly wage. John thus wants to base his decision about how many hours to work in each store on a different criterion: work stress factor. Based on interviews with present employees, John estimates that, on a scale of 1 to 10, the stress factors are 8 and 6 at stores 1 and 2, respectively. Because stress mounts by the hour, he presumes that the total stress at the end of the week is proportional to the number of hours he works in the store. How many hours should John work in each store?
5. OilCo is building a refinery to produce four products: diesel, gasoline, lubricants, and jet fuel. The demands (in bbl/day) for these products are 14,000, 30,000, 10,000, and 8,000, respectively. Iran and Dubai are under contract to ship crude to OilCo. Because of the production quotas specified by OPEC (Organization of Petroleum Exporting Countries), the new refinery can receive at least 40% of its crude from Iran and the remaining amount from Dubai. OilCo predicts that these demand and crude oil quotas will remain steady over the next 10 years.

   The different specifications of the two crude oils lead to different product mixes: One barrel of Iran crude yields .2 barrel of diesel, .25 barrel of gasoline, .1 barrel of lubricant, and .15 barrel of jet fuel. The corresponding yields from Dubai crude are .1, .6, .15, and .1, respectively.

   OilCo needs to determine the minimum capacity of the refinery (in barrels per day).
6. Day Trader wants to invest a sum of money that would generate an annual yield of at least $10,000. Two stock groups are available: blue chips and high tech, with average annual yields of 10% and 25%, respectively. Though high-tech stocks provide higher yield, they are more risky, and Trader wants to limit the amount invested in these stocks to no more than 60% of the total investment.

   What is the minimum amount Trader should invest in each stock group to accomplish the investment goal?

2.2.3 Graphical Solution with TORA

The design of TORA allows you to use the software in a tutorial or automated mode (or a combination of the two modes, if you desire). The software is menu-driven, and hence requires no user manual. Nevertheless, for your convenience, a TORA primer is given in Appendix C.

The graphical solution of LP problems by TORA requires the following steps:

1. Select `Linear Programming` from `MAIN menu`.
2. Specify input mode (existing file or new problem) and input format.

3. For new problems, use the input grid to enter data.
4. Press `Solve Menu`.
5. Select `Solve` ⇒ `Graphical` from the `SOLVE/MODIFY` menu.
6. Specify the output format and then press `GoTo Output Screen`.
7. Graph and solve the LP model.

Figure 2.4 shows the graphical solution of the Reddy Mikks model (file ch2ToraReddyMikks.txt). The algebraic LP is given in the left window of the screen. The right window starts with a blank first quadrant and properly scaled x_1 and x_2 axes, exactly similar to what you would be doing if you are graphing the problem on paper. You can graph the LP in two ways: If you click the bottom entry in the left window, the entire LP will be graphed in one stroke. Else, you may click the constraints one at a

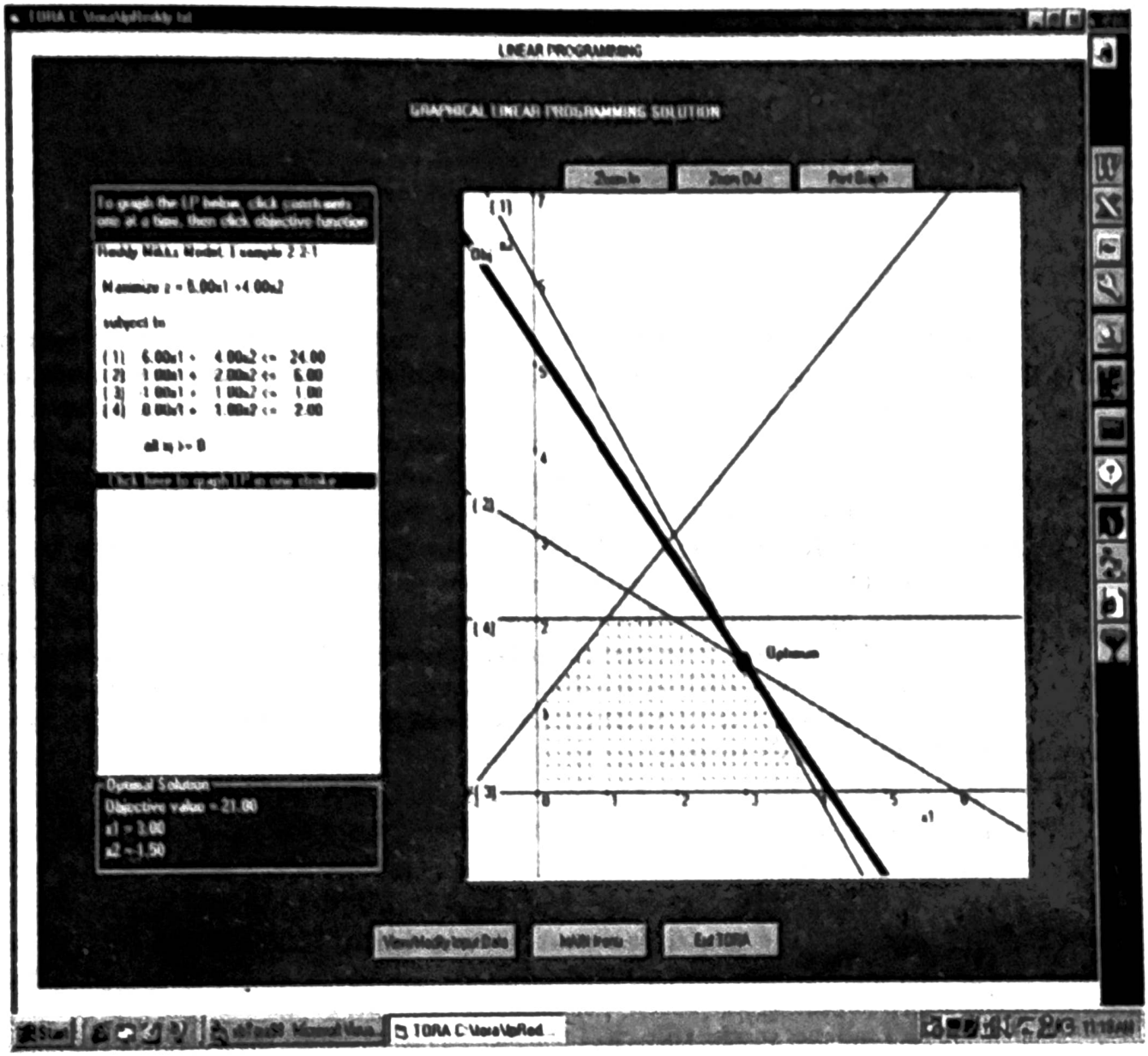


FIGURE 2.4
TORA graphical output of the Reddy Mikks model

time (in any order), and then click the objective function to produce an animated display of the determination of the optimum solution.

To allow greater flexibility in experimenting with TORA's graphical module, the entire graph may be restarted by clicking the nonnegativity constraint entry `all xj>= 0` in the left window. You may also modify the current LP by clicking the `View/Modify` and then resolve the new model.

PROBLEM SET 2.2C

1. *TORA Tutorial.* Input the following LP into TORA and select the graphic solution mode to reveal the LP graphic screen.

$$\text{Minimize } z = 3x_1 + 8x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\geq 8 \\ 2x_1 - 3x_2 &\leq 0 \\ x_1 + 2x_2 &\leq 30 \\ 3x_1 - x_2 &\geq 0 \\ x_1 &\leq 10 \\ x_2 &\geq 9 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Next, on a sheet of paper, graph and scale the x_1 and x_2 axes for the problem (you may also click `Print Graph` on the top of the right window to obtain a ready-to-use scaled sheet). Now, graph a constraint manually, and then click it on the left window of the screen to check your answer. Repeat the same for each constraint and then terminate the procedure with a graph of the objective function. The suggested process is designed to test and reinforce your understanding of the graphical LP solution through immediate feedback from TORA.

2. Consider the Reddy Mikks model (file ch2ToraReddyMikks.txt). Use TORA to demonstrate that the optimum LP solution is always associated with a corner point of the solution space. Specifically, use TORA's `View/Modify Input Data` option and the following objective functions to show that changes in the slope of the objective function could locate the optimum solution at a different corner point. The conclusion from this exercise is that corner points of the solution space are all that is needed to determine the optimum solution of the LP problem.
   (a) $z = 5x_1 + x_2$
   (b) $z = 5x_1 + 4x_2$
   (c) $z = x_1 + 3x_2$
   (d) $z = -x_1 + 2x_2$
   (e) $z = -2x_1 + x_2$
   (f) $z = -x_1 - x_2$
3. Consider the diet model (file ch2ToraDiet.txt) and change the objective function to

$$\text{Minimize } z = 8x_1 + .8x_2$$

Use TORA's graphical module to show that the optimum solution is associated with *two* distinct corner points and that both points yield the same objective value. In this case, the problem is said to have *alternative optima*. Explain the conditions leading to this situation and show that, in effect, the problem has infinity of alternative optima; then provide a formula for determining all such solutions.

4. Consider the following LP model:

$$\text{Maximize } z = 5x_1 + 4x_2$$

subject to

$$\begin{aligned} 6x_1 + 4x_2 &\le 24 \\ 6x_1 + 3x_2 &\le 22.5 \\ x_1 + x_2 &\le 5 \\ x_1 + 2x_2 &\le 6 \\ -x_1 + x_2 &\le 1 \\ x_2 &\le 2 \\ x_1, x_2 &\ge 0 \end{aligned}$$

In LP, a constraint is said to be *redundant* if its removal from the model leaves the feasible solution space unchanged. Use the graphical facility of TORA to identify the redundant constraints and then show that their removal does not affect the solution space or the optimal solution.

5. In the Reddy Mikks model, use TORA to show that the removal of the raw material constraints (constraints 1 and 2) will result in an unbounded solution space. What can be said in this case about the optimal solution of the model?

6. In the Reddy Mikks model, suppose that the following constraint is added to the problem.

$$x_2 \ge 3$$

Use TORA to show that the resulting model has conflicting constraints that cannot be satisfied simultaneously, and hence it has no feasible solution.

2.3 GRAPHICAL SENSITIVITY ANALYSIS

An LP model is a snapshot of a real situation in which the model parameters (objective and constraint coefficients) assume static values. To enhance the applicability of LP in practice, we need to add a dynamic dimension that investigates the impact of making changes in the model parameters (objective and constraint coefficients) on the optimal solution. The procedure is referred as **sensitivity analysis** because it studies the sensitivity of the optimal solution to changes made in the model.

This section investigates two cases of sensitivity analysis based on the graphical LP solution: (1) changes in the objective coefficients and (2) changes in the right-hand side of the constraints. Although the presentation is elementary and limited in scope, it does provide fundamental insights into the development of sensitivity analysis. A complete presentation of the topic is given in Chapter 4.

2.3.1 Changes in the Objective Function Coefficients

The general objective function in a two-variable LP problem can be written as

$$\text{Maximize or minimize } z = c_1x_1 + c_2x_2$$

Changes in the coefficients c_1 and c_2 will change the slope of z and, hence, possibly, the optimal corner point (see Figure 2.1 for an illustration). However, there is a range of variation for both c_1 and c_2 that will keep the current optimum unchanged. Specifically, we are interested in determining the *range of optimality* for the ratio $\frac{c_1}{c_2}$ (or $\frac{c_2}{c_1}$) that will keep the present optimum solution unchanged. The following example illustrates the procedure.

Example 2.3-1

Consider the Reddy Mikks model (Example 2.1-1). In Figure 2.5, the optimum solution at C provides the maximum value of $z = 5x_1 + 4x_2$. If we change the objective function to $z = c_1x_1 + c_2x_2$, then the solution at C will remain optimum so long as the slope of z lies between the slopes of the two lines intersecting at C—namely, $6x_1 + 4x_2 = 24$ (raw material, $M1$) and $x_1 + 2x_2 = 6$ (raw material, $M2$). We can express this relationship algebraically as

$$\text{If } c_1 \neq 0, \text{ then } \frac{4}{6} \leq \frac{c_2}{c_1} \leq \frac{2}{1}$$

or

$$\text{If } c_2 \neq 0, \text{ then } \frac{1}{2} \leq \frac{c_1}{c_2} \leq \frac{6}{4}$$

In the first condition, $c_1 \neq 0$ signifies that the objective function line cannot be horizontal. Similarly, in the second condition, $c_2 \neq 0$ means that z cannot be vertical. As can be seen from Figure 2.5, the optimality range in this model (defined by the two lines intersecting at C) does not permit the objective function $z = c_1x_1 + c_2x_2$ to be a horizontal or vertical line. As a result, either of the given two conditions is applicable to this example. For the cases where both c_1 and c_2 can assume zero values, the range for $\frac{c_1}{c_2}$ (or $\frac{c_2}{c_1}$) must be partitioned into two sets in which the denominators are not allowed to be zero. See Problem 1, Set 2.3a for illustrations.

What the conditions for $\frac{c_1}{c_2}$ and $\frac{c_2}{c_1}$ say is that so long as these ratios are within the specified limits, the optimum solution remains unchanged at C. Note that if $z = c_1x_1 + c_2x_2$ happens to coincide with $x_1 + 2x_2 = 6$, then alternative optima can occur anywhere on the line segment CD. Similarly, if it coincides with $6x_1 + 4x_2 = 24$, then all the points on the line segment BC provide alternative optima. This observation, nevertheless, does not change the fact that C remains optimum in both cases.

We can use the given conditions to determine the optimal range for one of the coefficients, given that the other remains as originally given in $z = 5x_1 + 4x_2$. Thus, given $c_2 = 4$, the associated optimal range for c_1 is determined from the condition $\frac{1}{2} \leq \frac{c_1}{c_2} \leq \frac{6}{4}$ by substituting $c_2 = 4$, which yields $4 \times \frac{1}{2} \leq c_1 \leq 4 \times \frac{6}{4}$ or $2 \leq c_1 \leq 6$. Similarly, given $c_1 = 5$, the condition $\frac{4}{6} \leq \frac{c_2}{c_1} \leq 2$ will yield $\frac{10}{3} \leq c_2 \leq 10$.

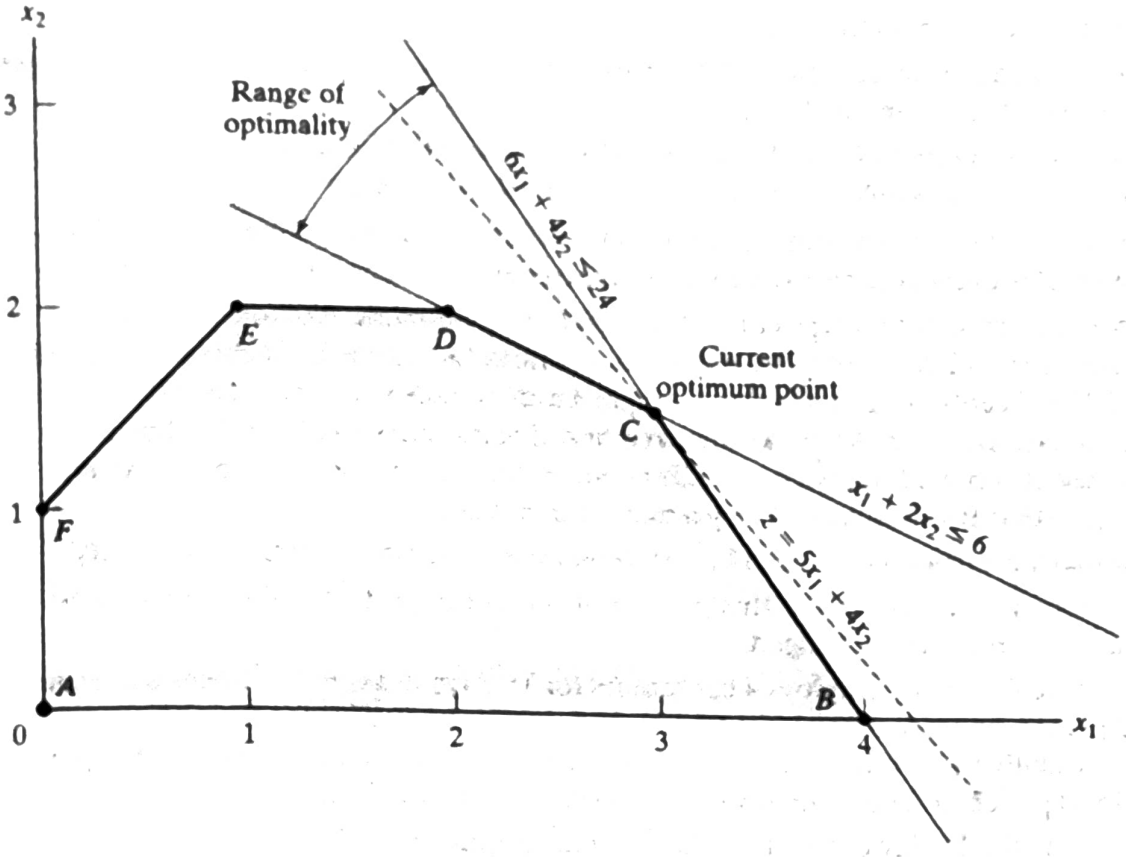


FIGURE 2.5
Range of optimality for the Reddy Mikks model

PROBLEM SET 2.3A

1. Determine graphically the range of optimality, $\frac{c_1}{c_2}$ or $\frac{c_2}{c_1}$ for the following problems. Observe the special cases where c_1 or c_2 may assume a zero value.

(a) Maximize $z = 2x_1 + 3x_2$
subject to

$$3x_1 + 2x_2 \le 6$$
$$-x_1 + x_2 \le 0$$
$$x_1, x_2 \ge 0$$

(b) Maximize $z = 6x_1 + 3x_2$
subject to

$$3x_1 + 2x_2 \le 6$$
$$x_1 - x_2 \le 0$$
$$x_1, x_2 \ge 0$$

(c) Maximize $z = x_1 + x_2$
subject to

$$-x_1 + x_2 \le 0$$
$$3x_1 - x_2 \le 3$$
$$x_1, x_2 \ge 0$$

2. In the diet problem of Example 2.2-2,
   (a) Determine the optimality range for the ratio of the cost per pound of corn to the cost per pound of soybean meal.
   (b) If the cost per pound of corn is increased by 20% and that of soybean meal is decreased by 5%, would the current solution remain optimal?
   (c) If the cost per pound of corn is increased to 70 cents and the cost per pound of soybean meal is decreased to 50 cents, would the current solution remain optimum?

3. B&K grocery store sells two types of soft drinks: the brand name A1 Cola and the cheaper store brand B&K Cola. The margin of profit for A1 is about 5 cents per can and that of B&K is 7 cents per can. On the average, the store sells no more than 500 cans of both colas a day. Although A1 is a recognized brand name, customers tend to buy more B&K because it is considerably cheaper. It is estimated that at least 100 cans of A1 are sold daily and that B&K outsells A1 by a margin of at least 2:1.
   (a) How many cans per day of each brand should the store carry to maximize profit?
   (b) Determine the ratio of the profits per can of A1 and B&K that will keep the optimum solution in (a) unchanged.

4. Baba Furniture Company employs 4 carpenters for 10 days to assemble tables and chairs. It takes 2 person-hours to assemble a table and .5 person-hour to assemble a chair. Customers usually buy a table and between four and six chairs. The profits are $135 per table and $50 per chair. The company operates one 8-hour shift a day.
   (a) Determine the 10-day optimal production mix graphically.
   (b) Determine the range of the ratio of unit profits that will keep the optimum in (a) unchanged.
   (c) If the present profits per table and chair are each reduced by 10%, use the answer in (b) to show how this change may affect the optimum solution obtained in (a).
   (d) If the present profits per table and chair are changed to $120 and $25, use sensitivity result in (b) to determine whether or not the solution in (a) will change.

5. The Bank of Elkins is allocating a maximum of $200,000 for personal and car loans during the next month. The bank charges 14% for personal loans and 12% for car loans. Both types of loans are repaid at the end of a 1-year period. Experience shows that about 3% of personal loans and 2% of car loans are never repaid, both as principal and interest. The bank usually allocates at least twice as much to car loans as to personal loans.
   (a) Determine the optimal allocation of funds between the two loans and the net rate of return on all the loans.
   (b) Determine the optimality range for the ratio of car to personal interest rates that will keep the solution in (a) unchanged.
   (c) Suppose that the percentages of unrepaid personal and car loans are changed to 4% and 3%, respectively; use the result in (b) to show if this change will affect the optimum solution in (a).

6. Electra produces two types of electric motors, each on a separate assembly line. The respective daily capacities of the two lines are 600 and 750 motors. Type 1 motor uses 10 units of a certain electronic component, and type 2 motor uses only 8 units. The supplier of the component can provide 8000 pieces a day. The profits per motor for types 1 and 2 are $60 and $40, respectively.
   (a) Determine the optimum daily production mix.
   (b) Determine the optimality range of the ratio of unit profits that will keep the solution in (a) unchanged.

7. Popeye Canning is contracted to receive 60,000 lb of ripe tomatoes at 7 cents per pound from which it produces both canned tomato juice and tomato paste. The canned products are packaged in 24-can cases. A can of juice uses 1 lb of fresh tomatoes, and a can of paste uses $\frac{1}{3}$ lb only. The company's share of the market is limited to 2000 cases of juice and 6000 cases of paste. The wholesale prices per case of juice and paste are \$18 and \$9, respectively.
   (a) Develop an optimum production program for Popeye.
   (b) Determine the price ratio of juice to paste that will allow Popeye to produce more cases of juice than of paste.
8. Dean's Furniture Company assembles from precut lumber two types of kitchen cabinets: regular and deluxe. The regular cabinets are painted white, and the deluxe are varnished. Both painting and varnishing are carried out in one department. The assembly department can produce a maximum of 200 regular cabinets and 150 deluxe per day. Varnishing a deluxe unit takes twice as much time as painting a regular one. If the painting/varnishing department is dedicated to the deluxe units only, it can complete 180 units daily. The company estimates that the profits per unit for the regular and deluxe cabinets are \$100 and \$140, respectively.
   (a) Formulate the problem as a linear program and find the optimal production schedule per day.
   (b) Suppose that because of competition, the profits per unit of the regular and deluxe units must be reduced to \$80 and \$110, respectively. Use sensitivity analysis to determine whether or not the optimum solution in (a) remains unchanged.

2.3.2 Change in Availability of Resources

In LP models, constraints, directly or indirectly, represent the usage of limited resources. In this case, the right-hand side can be thought of as representing limits on the availability of the resources. This section investigates the sensitivity of the optimum solution to making changes in the amount of available resources.

Example 2.3-2

Consider the Reddy Mikks model. Figure 2.6 shows that the current optimum occurs at C is the intersection of the lines associated with raw materials $M1$ and $M2$. When the availability of $M1$ changes (increases or decreases around its current level of 24 tons), and given $M2 = 6$ tons, the optimum solution at point C will "slide" along the line segment DG. Any change in $M1$ outside the range of this segment will render point C (intersection of the lines associated with $M1$ and $M2$) infeasible. For this reason, we say that the end points $D = (2, 2)$ and $G = (6, 0)$ delineate the *feasibility range* for $M1$. Thus,

$$\text{Amount of } M1 \text{ at } D = 6x_1 + 4x_2 = 6 \times 2 + 4 \times 2 = 20 \text{ tons}$$

$$\text{Amount of } M1 \text{ at } G = 6x_1 + 4x_2 = 6 \times 6 + 4 \times 0 = 36 \text{ tons}$$

It then follows that, given $M2 = 6$, the feasibility range for $M1$ is

$$20 \le M1 \le 36$$

The result shows that $M1$ can be decreased by as much as 4 tons or increased by as much as 12 tons while guaranteeing that the optimum solution point will be given by

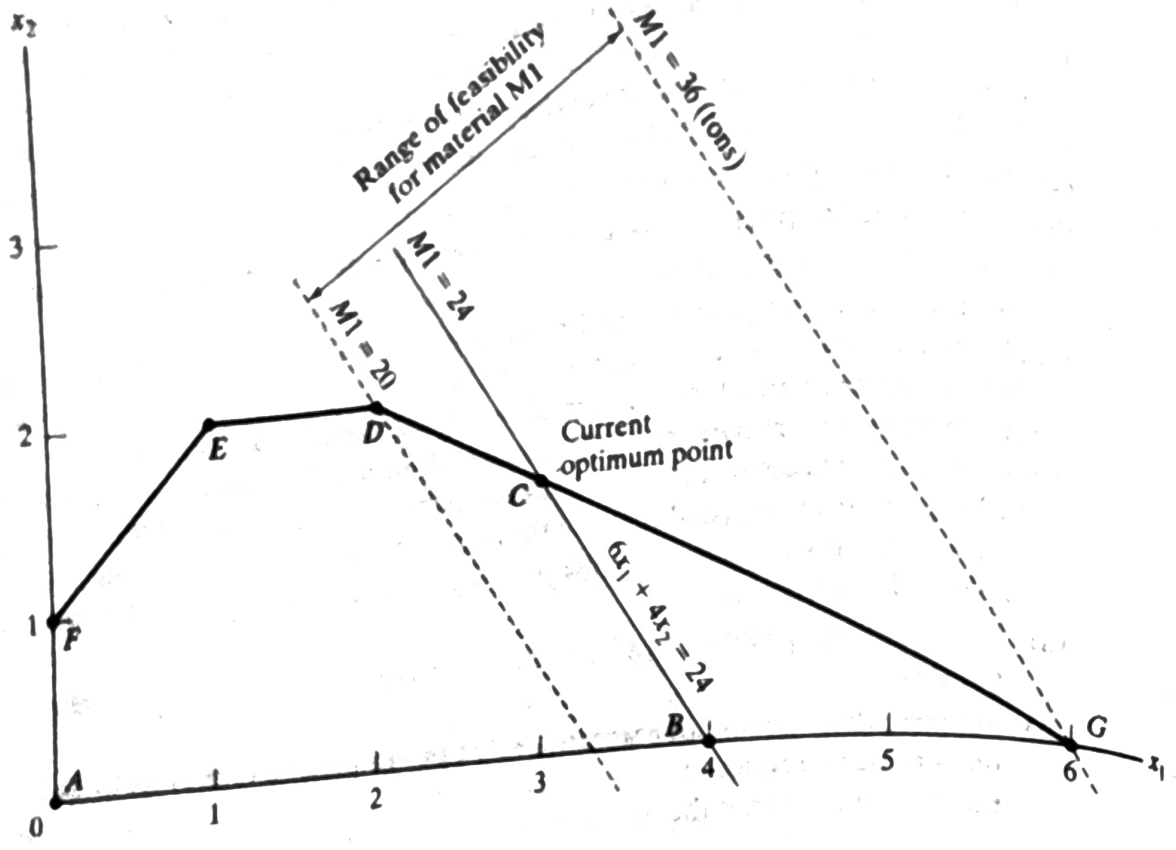


FIGURE 2.6
Range of feasibility for raw material $M1$ in the Reddy Mikks model

the intersection of the lines associated with $M1$ and $M2$. Indeed, given $M_2 = 6$, the associated general solution is obtained in terms of $M1$ as (verify!)

$$\left.\begin{aligned} x_1 &= \tfrac{1}{4}M1 - 3 \\ x_2 &= -\tfrac{1}{8}M1 + \tfrac{9}{2} \end{aligned}\right\}, 20 \le M1 \le 36$$

Next, consider raw material $M2$. Figure 2.7 shows that the feasibility range for $M2$ (given $M1 = 24$ tons) is delineated by the end points B and H, where $B = (4,0)$ and $H = (\frac{8}{3}, 2)$, where point H is defined by the intersection of lines ED and BC. Thus,

$$\text{Amount of } M2 \text{ at } B = x_1 + 2x_2 = 4 + 2 \times 0 = 4 \text{ tons}$$

$$\text{Amount of } M2 \text{ at } H = \tfrac{8}{3} + 2 \times 2 = \tfrac{20}{3} \text{ tons}$$

Thus, given $M1 = 24$, the feasibility range for $M2$ is

$$4 \le M2 \le \tfrac{20}{3}$$

Again, you can verify that, given $M1 = 24$, the associated solution is given as

$$\left.\begin{aligned} x_1 &= -\tfrac{1}{2}M2 + 6 \\ x_2 &= \tfrac{3}{4}M2 - 3 \end{aligned}\right\}, 4 \le M2 \le \tfrac{20}{3}$$

2.3.3 Unit Worth of a Resource

Figure 2.8 shows that an LP model can be thought of as an input-output model in which the limited resources represent the input and the value of the objective function

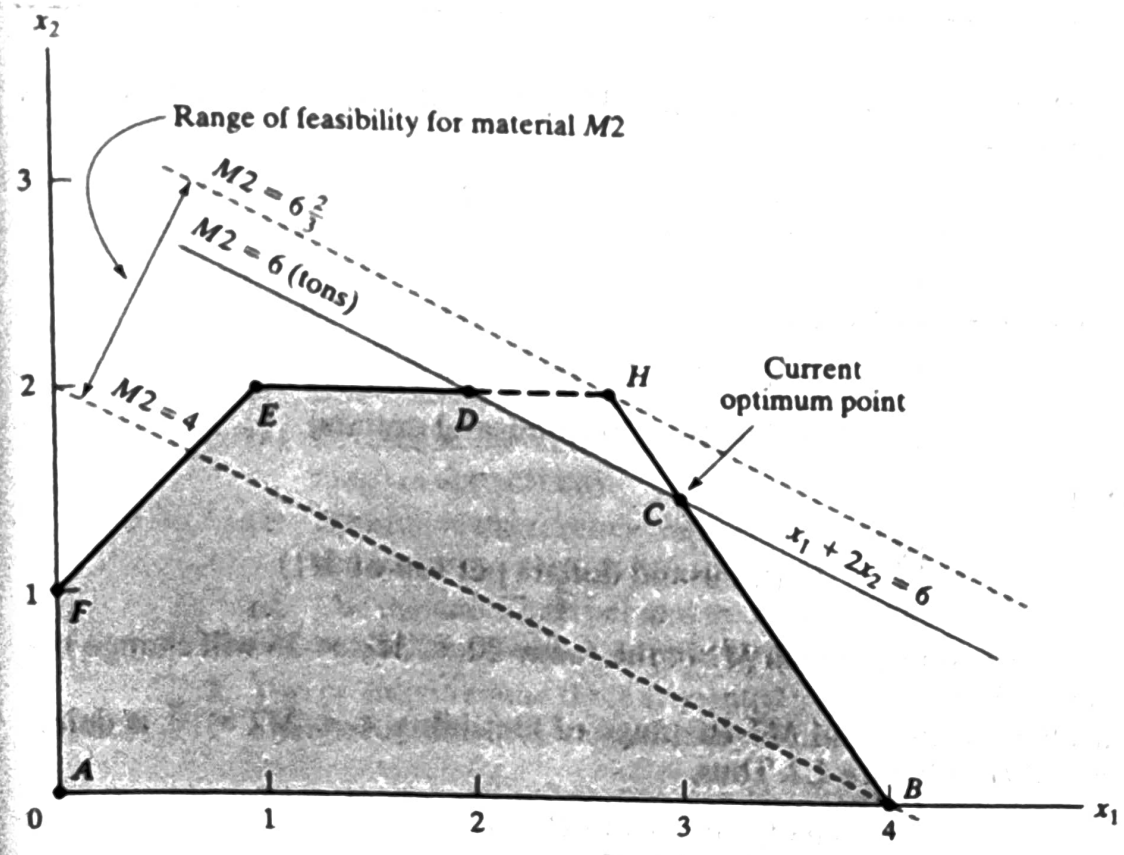


FIGURE 2.7
Range of feasibility for raw material $M2$ in the Reddy Mikks model

summarizes the output. A useful measure of the model is to determine how changes in its input (resources) can impact its output (the objective value). Such a measure can be obtained as a by-product of the calculation of the feasibility range presented in Section 2.3.2. Specifically, we seek to determine the **unit worth of a resource**, which is defined as the rate of change in the optimum objective value that results from making changes in the available amount of a resource.

Letting y_i represent the worth per unit of resource i, the associated formula for computing this measure is

$$y_i = \frac{\text{change in value of } z \text{ corresponding to the feasible range of resource } i}{\text{feasible range of resource } i}$$

The Reddy Mikks model is used to illustrate the new measure.

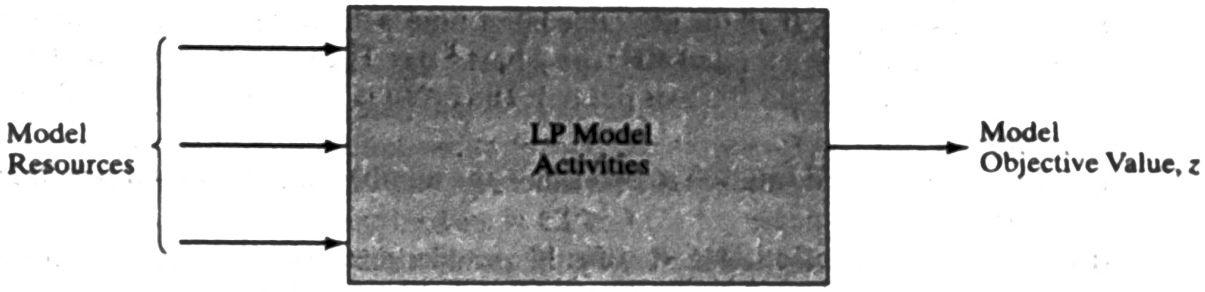


FIGURE 2.8
Representation of LP as an input-output model

Example 2.3-3

Figure 2.6 shows that the feasible range for $M1$, $20 \le M1 \le 36$, is delineated by points D and G. Thus, we have

$$y_1 = \frac{\text{change in } z \text{ from } D \text{ to } G}{\text{change in } M1 \text{ from } D \text{ to } G}$$

Given $D = (2, 2)$ and $G = (6, 0)$, then

$$z \text{ at } D = 5 \times 2 + 4 \times 2 = 18 \text{ (thousand dollars)}$$
$$z \text{ at } G = 5 \times 6 + 4 \times 0 = 30 \text{ (thousand dollars)}$$

It then follows that

$$y_1 = \frac{30 - 18}{36 - 20} = .75 \text{ (thousand dollars per ton of } M1)$$

The result shows that a 1-ton change in $M1$ in the range $20 \le M1 \le 36$ will change the optimum value of z by \$750.

Next, we consider raw material $M2$. Its range of feasibility, $4 \le M2 \le \frac{20}{3}$, is delineated by points B and H in Figure 2.7. Thus,

$$y_2 = \frac{\text{change in } z \text{ from } B \text{ to } H}{\text{change in } M2 \text{ from } B \text{ to } H}$$

where

$$z \text{ at } B = 5x_1 + 4x_2 = 5 \times 4 + 4 \times 0 = 20 \text{ (thousand dollars)}$$
$$z \text{ at } H = 5 \times \tfrac{8}{3} + 4 \times 2 = \tfrac{64}{3} \text{ (thousand dollars)}$$

It then follows that

$$y_2 = \frac{\frac{64}{3} - 20}{\frac{20}{3} - 4} = .5 \text{ (thousand dollar per ton of } M2)$$

Once again this result says that an increase (decrease) in $M1$ by one ton in the range $4 \le M1 \le \frac{20}{3}$ increases (decreases) the profit by \$500.

PROBLEM SET 2.3B

1. Wild West produces two types of cowboy hats. Type 1 hat requires twice as much labor time as a type 2. If all the available labor time is dedicated to type 2 alone, the company can produce a total of 400 type 2 hats a day. The respective market limits for the two types are 150 and 200 hats per day. The profit is \$8 per type 1 hat and \$5 per type 2 hat.
   (a) Use the graphical solution to determine the number of hats of each type that would maximize profit.
   (b) Determine the worth of increasing the production capacity of the company by one type 2 hat and the range for which this result is applicable.

(c) If the daily demand limit on type 1 hat is decreased to 120, use the unit worth of the resource to determine the corresponding effect on the optimal profit.

(d) What is the worth per unit increase in the market share of type 2 hat? By how much can the market share be increased while yielding the computed worth per unit?

2. A company produces two products, A and B. The sales volume for A is at least 80% of the total sales of both A and B. However, the company cannot sell more than 100 units of A per day. Both products use one raw material, of which the maximum daily availability is 240 lb. The usage rates of the raw material are 2 lb per unit of A and 4 lb per unit of B. The unit prices for A and B are \$20 and \$50, respectively.

(a) Determine the optimal product mix for the company.

(b) Determine the worth per unit change in the availability of the raw material and its range of applicability.

(c) Determine the revenue range corresponding to the feasibility range of the raw material resource.

(d) Use the unit worth per unit to determine the effect of changing the maximum demand for product A by $\pm$ 10 units.

3. Two products require three sequential processes. The time available for each process is 10 hours a day. The following table summarizes the data of the problem:

	Minutes per unit			
Product	*Process 1*	*Process 2*	*Process 3*	*Unit profit*
1	10	6	8	\$2
2	5	20	10	\$3

(a) Determine the optimal mix of the two products.

(b) Devise a procedure for prioritizing the three processes for possible expansion.

4. Show & Sell can advertise its products on local radio or television (TV). The advertisement budget is limited to \$10,000 a month. Each minute of radio advertisement costs \$15 and each minute of TV commercials costs \$300. Show & Sell likes to use radio advertisement at least twice as much as TV. In the meantime, it is not practical to use more than 400 minutes of radio advertisement a month. Past experience shows that TV advertisement is estimated to be 25 times more effective than that of radio.

(a) Determine the optimum allocation of the budget to radio and TV advertisements.

(b) Determine the worth per unit of increasing the monthly limit on radio advertisement.

(c) If the monthly budget is increased to \$15,000, use the unit worth measure to determine the resulting optimum measure of advertisement effectiveness.

5. Wyoming Electric Co-op owns a steam-turbine power-generating plant. Because Wyoming is rich in coal deposits, the plant generates its steam from coal. This, however, may result in emission that does not meet Environmental Protection Agency standards. EPA regulations limit sulfur dioxide discharge to 2000 parts per million per ton of coal burned and smoke discharge from the plant stacks to 20 lb per hour. The Co-op receives two grades of pulverized coals, $C1$ and $C2$, for use in the steam plant. The two grades are usually mixed together before burning. For simplicity, it can be assumed that the sulfur pollutant of the mixture (in parts per million) is a weighted average of the proportion of